Sri Lanka Institute of Information Technology

IT2130 – Operating Systems and System Administration
Year 2, Semester 2- 2026

Practical Answer Submission

Practical Sheet No: 08

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Question 1

Consider the program below and explain the code line by line

```
  GNU nano 7.2                                                  d1.c
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <string.h>

int main(){
 key_t key=1234;
 int shmid;
 char*ptr;

 shmid=shmget(key,4096,0666 | IPC_CREAT);

 ptr=(char*) shmat(shmid,NULL,0);

 sprintf(ptr,"Hello from shared memory!");

 printf("Data written to shared memory\n");

 shmdt(ptr);

 return 0;

}
```

```
kamshiga@kamshiga-VirtualBox:~$ nano d1.c
kamshiga@kamshiga-VirtualBox:~$ gcc -o d1 d1.c
kamshiga@kamshiga-VirtualBox:~$ ./d1
Data written to shared memory
kamshiga@kamshiga-VirtualBox:~$
```

1 **#include <stdio.h>** - Used for Input Output Functions

2 **#include <sys/ipc.h>** - Provides definitions for IPC(Inter Process Communication)

3 **#include <sys/shm.h>** - Used for shared memory functions

4 **#include <string.h>** - Used for string handling functions

5 **int main()** - Main function where program execution starts

6 **key_t key=1234** - Unique key used to identify the shared memory segment

7 **int shmid** - Variable to store shared memory ID

8 **char *ptr** - Pointer to access shared memory

9 **shmid = shmget(key, 4096, 0666 | IPC_CREAT**)

- Creates a shared memory segment of size 4096 bytes. The permission 0666 allows read and write access, and IPC_CREAT ensures that the memory is created if it does not already exist.

10 **ptr = (char *) shmat(shmid, NULL, 0)** - Attaches shared memory to the process and returns pointer

11 **sprintf(ptr, "Hello from shared memory!")** - Writes message into shared memory location

12 **printf("Data written to shared memory\n")** - Displays confirmation message

13 **shmdt(ptr)** - Detaches shared memory from process

14 **return 0** - Ends program successfully

Question 02

Consider the program below and explain the code line by line

```
GNU nano 7.2                                   d2.c
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <string.h>

int main() {
 key_t key =1234;
 int shmid;
 char * ptr;

 shmid=shmget(key,4096, 0666);

 ptr=(char *)shmat(shmid,NULL,0);

 printf("Read from shared memory:%s\n",ptr);

 shmdt(ptr);

 shmctl(shmid,IPC_RMID,NULL);

 return 0;
}
```

```
kamshiga@kamshiga-VirtualBox:~$ nano d2.c
kamshiga@kamshiga-VirtualBox:~$ gcc -o d2 d2.c
kamshiga@kamshiga-VirtualBox:~$ ./d2
Read from shared memory:Hello from shared memory!
kamshiga@kamshiga-VirtualBox:~$
```

1 **#include <stdio.h>**

2 **#include <sys/ipc.h>**

3 **#include <sys/shm.h>**

-Libraries for I/O and shared memory

4 **int main()** – Main function where program execution starts

5 **key_t key = 1234** - Unique key used to identify the shared memory segment

6 **int shmid** - Variable to store shared memory ID

7 **char *ptr** - Pointer to access shared memory

8 **shmid = shmget(key, 4096, 0666)** - Access the existing shared memory segment. Unlike the previous program, it does not use IPC_CREAT, because the memory is already created.

9 **ptr = (char *) shmat(shmid, NULL, 0)** - Attach shared memory to the process and returns a pointer to it.

10 **printf("Read from shared memory: %s\n", ptr)** - Reads and prints data from shared memory

11 **shmdt(ptr)** - Detach shared memory

12 **shmctl(shmid, IPC_RMID, NULL)** - Deletes the shared memory segment

13 **return 0** - Ends program successfully

Question 3

Write a C program to implement a shared memory using system calls. Create a shared memory segment and write a message such as "Operating Systems Lab" into it. Then read the message from the shared memory and print it to the screen.

```
kamshiga@kamshiga-VirtualBox: ~                    kamshiga@kamshig
 GNU nano 7.2                                   d3.c
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <string.h>

int main(){
 key_t key = 1234;
 int shmid;
 char *ptr;

 shmid=shmget(key, 4096, 0666 | IPC_CREAT);

 ptr =(char*) shmat(shmid,NULL, 0);

 strcpy(ptr,"Operating Systems Lab");
 printf("Message written to shared memory\n");

 printf("Message read from shared memory: %s\n", ptr);

 shmdt(ptr);

 shmctl(shmid, IPC_RMID, NULL);

 return 0;

}
                                        [ Read 26 lines ]
```

```
kamshiga@kamshiga-VirtualBox: ~                    kamshiga@kamshig
kamshiga@kamshiga-VirtualBox:~$ nano d3.c
kamshiga@kamshiga-VirtualBox:~$ gcc -o d3 d3.c
kamshiga@kamshiga-VirtualBox:~$ ./d3
Message written to shared memory
Message read from shared memory: Operating Systems Lab
kamshiga@kamshiga-VirtualBox:~$
```